

The Switch Feature Handler: The in the SDN is responsible for managing the initial connection between the SDN controller and a switch. Its primary purpose is to ensure that the switch can handle unmatched packets by installing a table-miss flow entry, which directs such packets to the controller for further processing.

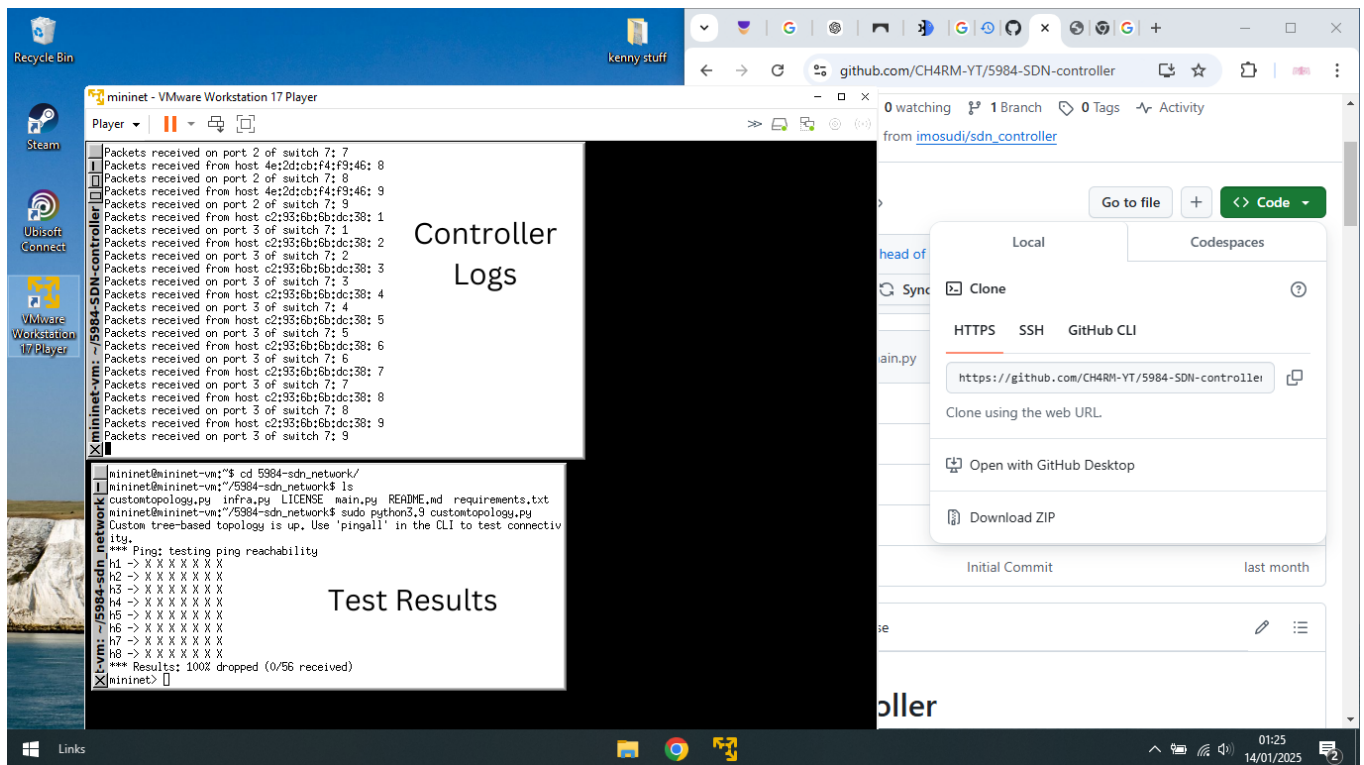
Table-Miss Flow Installation: Table-miss flows are essential in SDN to handle packets that do not match any predefined flow rules, ensuring they are not dropped by default. They act as a catch-all rule with the lowest priority (`priority=0`) and typically forward unmatched packets to the controller using an action like `OFActionOutput(OFPP_CONTROLLER)`. This allows the controller to analyze traffic, make decisions dynamically (e.g., install new flow rules), and maintain centralized control. Table-miss flows also enable network visibility for monitoring, logging, and troubleshooting, supporting flexible and dynamic network management.

Packet-In Handler: When a packet is sent to the SDN controller via a Packet-In message, the controller's event handler processes it by analyzing its headers, metadata (e.g., input port), and payload. This enables the controller to determine the appropriate action, such as installing new flow rules on the switch, modifying the packet, or forwarding it to a specific destination. The controller may also log or monitor the packet for policy enforcement, traffic analysis, or debugging purposes before sending back a decision through a Flow Mod or Packet-Out message.

Performance Analysis

In the custom tree-based topology, packets flow from hosts to access switches, then to aggregation switches, and finally to the core switch. However, the packets flow in such a manner only if the source and destination IP addresses fall within the defined subnets (`10.0.0.0/24`, `10.0.1.0/24`), but in the case of this custom topology. When attempting communication with the controller having an IP address that is not in the specified subnet (`192.168.233.129`), the packets are dropped by the controller due to its subnet filtering logic. As a result, only packets within the specified subnets are successfully forwarded across the network, while packets involving the controller are intentionally discarded.

PLEASE ZOOM IN FOR A CLEARER PICTURE



The above shows the logs for the controller and the test result of the “pingall” test that was ran on the mininet CLI

The controller logs show that the packets were received successfully from the host and shows the respective port the packet was received on

The Test Result show that 100% of the packets were dropped.